

Lab 1: Film Ranking (The Basics)

The screenshot shows a PostgreSQL client interface with the following components:

- Query Editor:** Contains a SQL query that uses a window function to rank films by length within each category.
- Query:**

```
-- )
-- SELECT * FROM RankedCust WHERE rn = 1;

WITH RankedFilms AS (
  SELECT c.name AS category, f.title AS film_title, f.length AS film_len,
         DENSE_RANK() OVER (PARTITION BY c.name ORDER BY f.length DESC, f.title) AS rn
  FROM film f
  JOIN film_category fc ON f.film_id = fc.film_id
  JOIN category c ON c.category_id = fc.category_id
)
SELECT * FROM RankedFilms WHERE rn <= 3
```
- Data Output:** Displays the results of the query in a table format.

Data Output Table:

	category character varying (25)	film_title character varying (255)	film_len smallint	rn bigint
1	Action	Darn Forrester	185	1
2	Action	Worst Banger	185	2
3	Action	Casualties Encino	179	3
4	Animation	Gangs Pride	185	1
5	Animation	Pond Seattle	185	2
6	Animation	Sons Interview	184	3
7	Children	Fury Murder	178	1
8	Children	Wrong Behavior	178	2
9	Children	Empire Malkovich	177	3
10	Classics	Conspiracy Spirit	184	1
11	Classics	Dracula Crystal	176	2
12	Classics	Lights Deer	174	3
13	Comedy	Control Anthem	185	1
14	Comedy	Saturn Name	182	2
15	Comedy	Searchers Wait	182	3

Lab 2: Growth Analysis (Time Series)

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No limit

Query
Query History

```

53
54 WITH MonthlyRevenue AS (
55     SELECT
56         DATE_TRUNC('month', payment_date) AS month,
57         SUM(amount) AS current_revenue
58     FROM payment
59     GROUP BY DATE_TRUNC('month', payment_date)
60 ),
61 RevenueWithLag AS (
62     SELECT
63         month,
64         current_revenue,
65         LAG(current_revenue) OVER (ORDER BY month) AS previous_revenue
66     FROM MonthlyRevenue
67 )
68 SELECT
69     month,
70     current_revenue,
71     previous_revenue,
72     ROUND(
73         ((current_revenue - previous_revenue) / previous_revenue) * 100,
74         2
75     ) AS growth_percentage
76 FROM RevenueWithLag
77 ORDER BY month;
```

Data Output
Messages
Notifications

Show SQL

Showing rows: 1 to 4 Page No: 1 of 1

	month timestamp without time zone	current_revenue numeric	previous_revenue numeric	growth_percentage numeric
1	2007-02-01 00:00:00	8351.84	[null]	[null]
2	2007-03-01 00:00:00	23886.56	8351.84	186.00
3	2007-04-01 00:00:00	28559.46	23886.56	19.56
4	2007-05-01 00:00:00	514.18	28559.46	-98.20

Lab 3: Customer Lifetime Value (Running Total)


```

JOIN customer c ON p.customer_id = c.customer_id
JOIN address a ON c.address_id = a.address_id
JOIN city ci ON a.city_id = ci.city_id
JOIN country co ON ci.country_id = co.country_id
GROUP BY co.country, c.customer_id, c.first_name, c.last_name
),
RankedCustomers AS (
  SELECT
    country,
    customer_id,
    customer_name,
    total_spending,
    RANK() OVER (
      PARTITION BY country
      ORDER BY total_spending DESC
    ) AS country_rank
  FROM CustomerSpending
)
SELECT
  country,
  customer_id,
  customer_name,
  total_spending
FROM RankedCustomers
WHERE country_rank = 1
ORDER BY country;

```

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Query Query History ↗

```

93
94 WITH CustomerSpending AS (
95     SELECT
96         co.country,
97         c.customer_id,
98         c.first_name || ' ' || c.last_name AS customer_name,
99         SUM(p.amount) AS total_spending
100 FROM payment p
          
```

Data Output Messages Notifications ↗

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▼

SQL

Showing rows: 1 to 108 ✎ Page No: 1 of 1 ◀ ⏪ ⏩ ▶

	country character varying (50)	customer_id integer	customer_name text	total_spending numeric
1	Afghanistan	218	Vera Mccoy	67.82
2	Algeria	176	June Carroll	151.68
3	American Samoa	320	Anthony Schwab	47.85
4	Angola	528	Claude Herzog	93.80
5	Anguilla	381	Bobby Boudreau	99.68
6	Argentina	560	Jordan Archuleta	129.71
7	Armenia	41	Stephanie Mitch...	118.75
8	Austria	266	Nora Herrera	113.74
9	Azerbaijan	334	Raymond Mcwh...	108.75
10	Bahrain	590	Seth Hannon	108.76
11	Bangladesh	21	Michelle Clark	146.68
12	Belarus	144	Clara Shaw	189.60
13	Bolivia	431	Joel Francisco	92.78
14	Brazil	178	Marion Snyder	194.61
15	Brunei	91	Lois Butler	107.66
16	Bulgaria	540	Tyrone Asher	112.76
17	Cambodia	516	Elmer Noe	92.76
18	Cameroon	352	Albert Crouse	96.78
19	Canada	410	Curtis Irby	167.62
20	Chad	535	Javier Elrod	122.72

Total rows: 108
Query complete 00:00:02.189
LF Ln 126, Col 18